

SCH3U Percent Composition and Mole Quiz

CLASS SET

Completion

Complete each statement.

1. The total number of atoms present in 1 mol of gold is _____.
2. 6.02×10^{23} particles of nitrogen gas have a mass of _____.

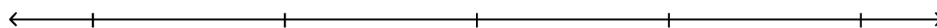
Short Answer

Show your work. Write a sentence stating the correct value for the following questions.

3. Sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, can be used as a fixer in the field of photography. Calculate the percentage composition, by mass, of each element in sodium thiosulfate.
4. A 15.00 g sample of $(\text{NH}_4)_2\text{SO}_4$ was analyzed and found to contain 2.28g of nitrogen.
 - a) Calculate the percent nitrogen in the sample.
 - b) Calculate the theoretical percentage of nitrogen in $(\text{NH}_4)_2\text{SO}_4$.
5. Convert 6.27 mol hydrogen peroxide, H_2O_2 , into mass (in grams).
6. Determine the number of carbon atoms found in 2.50 mol of methane, CH_4 . **Explain** your answer.
7. Calculate the number of moles of lead present in 8.6×10^{17} atoms of Pb.
8. Calculate the number of moles found in 168.0 g of iron(II) oxide.

MUST DO THIS: **DRAW THIS LINE** on your answer page and indicate how confident you feel in this topic and the accuracy of your quiz answers.

****After reviewing the answers & making your corrections, make a second mark on the line in your correction colour.**



Not Yet

Getting there

Got it!!

SCH3U Percent Composition and Mole Quiz
Answer Section

/23

BIG NOTE: If any of your answers have different exponents in scientific notation be sure you can get the correct answer on your calculator (eg answer is 3.0×10^{-6} but your wrong answer is 3.0×10^{-32} , you likely have set up the equation properly but not entered the scientific notation properly in your calculator.)

COMPLETION

1. ANS: 6.02×10^{23}

PTS: 1

REF: K/U

OBJ: 2.1

STA: CC1.01

2. ANS: 28.0 g

PTS: 1

REF: I

OBJ: 2.1

STA: CC1.01

PROBLEM

3. ANS:

$$m_{\text{Na}} = 23.00 \text{ u} \times 2 \text{ atoms} = 46.00 \text{ u}$$

$$m_{\text{S}} = 32.10 \text{ u} \times 2 \text{ atoms} = 64.20 \text{ u}$$

$$m_{\text{O}} = 16.00 \text{ u} \times 3 \text{ atoms} = 48.00 \text{ u}$$

$$m_{\text{total}} = 158.20 \text{ u}$$

$$\begin{aligned} \% \text{Na} &= \frac{46.00 \text{ u}}{158.20 \text{ u}} \times 100\% \\ &= 29.08\% \end{aligned}$$

$$\begin{aligned} \% \text{S} &= \frac{64.20 \text{ u}}{158.20 \text{ u}} \times 100\% \\ &= 40.58\% \end{aligned}$$

$$\begin{aligned} \% \text{O} &= \frac{48.00 \text{ u}}{158.20 \text{ u}} \times 100\% \\ &= 30.34\% \end{aligned}$$

The percentage composition, by mass, of sodium thiosulfate is 29.08% sodium, 40.58% sulfur, and 30.34% oxygen.

PTS: 1

REF: I

OBJ: 2.3

STA: CC2.04

4. ANS:
 a) $2.28/15.00 \times 100 = 15.2\%$
 b) $MM = 132.16u$, $Mn = 28.02u$
 $\% = 28.02/132.16 \times 100$
 $= 21.2\%$

PTS: 1

SHORT ANSWER

5. ANS:

This question uses $n = m/MM$ equation, rearranged to solve for mass below.

$$n = 6.27 \text{ mol}$$

$$M = 34.02 \text{ g/mol}$$

$$m = 6.27 \text{ mol} \times \frac{34.02 \text{ g}}{1 \text{ mol}}$$

$$= 2.13 \times 10^2 \text{ g}$$

The mass of 6.27 mol hydrogen peroxide is $2.13 \times 10^2 \text{ g}$. (NOTE if you move the decimal point to get out of scientific notation, your answer would be 213 g, SAME number)

PTS: 1

REF: I

OBJ: 2.2

STA: CC2.05

6. ANS:

$$N_{\text{CH}_4} = nN_A$$

$$= 2.50 \text{ mol} \times \frac{6.02 \times 10^{23} \text{ molecules}}{1 \text{ mol}}$$

$$= 1.51 \times 10^{24} \text{ molecules}$$

Since there is one carbon atom in each methane molecule, there are 1.51×10^{24} atoms of carbon present.

PTS: 1

REF: I

OBJ: 4.4

STA: QC2.03

7. ANS:

This question uses $n = N/N_A$ equation.

$$n_{\text{Pb}} = 8.6 \times 10^{17} \text{ atoms} \times \frac{1.0 \text{ mol}}{6.02 \times 10^{23} \text{ atoms}}$$

$$= 1.4 \times 10^{-6} \text{ mol}$$

PTS: 1

REF: I

OBJ: 4.4

STA: QC2.03

8. ANS:

$$m = 168.0 \text{ g}$$

$$M = (1 \times 55.85) + (1 \times 16.00)$$

$$= 71.85 \text{ g/mol}$$

$$n_{\text{FeO}} = 168.0 \text{ g} \times \frac{1 \text{ mol}}{71.85 \text{ g}}$$

$$= 2.338 \text{ mol}$$

A mass of 168.0 g of iron(II) oxide is equivalent to 2.338 mol.

PTS: 1

REF: I

OBJ: 4.4

STA: QC2.03